

Geometry of Fractal Dimensional Dynamical Systems

1 Measure Theory

From "Fractal Geometry: Mathematical Foundations and Applications"

Definition. We call μ a *measure* on $\mathbb{R}^n$ if μ assigns a non-negative number, possibly ∞ , to each subset of $\mathbb{R}^n$ such that:

- (a) $\mu(\emptyset) = 0$
- (b) $\mu(A) \leq \mu(B)$ if $A \subset B$
- (c) if $A_1, A_2, \dots$ is a countable (or finite) sequence of sets then

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mu(A_i)$$

with equality if A_i are disjoint Borel sets.

Definition. If the n -dimensional parallelepiped defined as $P = \{(x_1, \dots, x_n) \in \mathbb{R}^n : (a_i \leq x_i \leq b_i)\}$ has a volume measure:

$$\text{vol}^n(P) = \prod_{i=1}^n (b_i - a_i)$$

Then we can extend the definition of volume to arbitrary sets by stitching together infinitesimal parallelepipeds. This is called the *n-dimensional Lebesgue measure* and is formally defined as follows:

$$\text{vol}^n(A) = \inf \left\{ \sum_{i=1}^{\infty} \text{vol}^n(P_i) : A \subset \bigcup_{i=1}^{\infty} P_i \right\}$$

Definition. Integration is best defined as a pairing between a measure μ and a scalar function $\varphi : \mathcal{X} \rightarrow \mathbb{R}$, with support $\mathcal{X} \subset \mathbb{R}$. First, we define a *simple* function $\hat{\varphi} : \mathcal{X} \rightarrow \{a_1, \dots, a_k\} \subset \mathbb{R}$, then we define the *Lebesgue integral* as the sum over level sets of $\hat{\varphi}$:

$$\int_{\mathcal{X}} \hat{\varphi} \, d\mu = \sum_{i=1}^k a_i \mu(\{x : \hat{\varphi}(x) = a_i\})$$

For a general smooth function φ , we define the Lebesgue integral as the integral of its best discrete approximation $\hat{\varphi}$. This definition should intuitively match how you learned integration in Calculus I except we've flipped the x and y axes so that the tiny rectangles go horizontally instead of vertically.

$$\int_{\mathcal{X}} \varphi d\mu = \sup \{ \langle \hat{\varphi}, \mu \rangle : \hat{\varphi} \text{ is simple, } 0 \leq \hat{\varphi} \leq \varphi \}$$

The definition above is the definite integral over $\mathcal{X} \subseteq \mathbb{R}^n$. For Borel subsets A of $\mathbb{R}^n$, we simply multiply φ (or equivalently, the measure) by the indicator function:

$$\int_A \varphi d\mu = \langle \varphi \mathbf{1}_A, \mu \rangle, \quad \mathbf{1}_A(x) = \begin{cases} 1, & x \in A \\ 0, & x \notin A \end{cases}$$

Definition. You may recognize that a measure also has a probabilistic interpretation, and our definition of integrals is exactly the expectation of an observable. To make this analogy rigorous, we define a *probability measure* to be a measure μ that satisfies $\mu(\Omega) = 1$ for some set Ω we call the *sample space*. It then follows from the definition of a measure that for any event $\mathcal{E} \subset \Omega$, the probability is bounded $0 \leq \mu(\mathcal{E}) \leq 1$. From this, you can go on to build all of probability theory. Joint distributions are just $\mu(A \cap B)$, conditionals are $\mu(A \cap B) / \mu(B)$, mean is just the inner product with the identity function $\langle \text{id}, \mu \rangle$, etc...

Definition. A probability measure μ is said to be absolutely continuous w.r.t. the Lebesgue measure if it has an associated *probability density function* $\rho(x)$ satisfying:

$$\int_A \varphi d\mu = \int_A \varphi(x) \rho(x) d \text{vol}^n$$

for all smooth functions $\varphi(x) : \mathcal{X} \rightarrow \mathbb{R}$. This integral is called the *expectation* of φ w.r.t. the probability measure μ and is commonly denoted $\mathbb{E}_{X \sim \mu}[\varphi(X)]$.

Note: the inner product notation is preferable because the symbol d is very confusing, especially when we introduce the exterior derivative operator in differential geometry (it is also denoted “ d ” but evaluates to 0 on top forms so $d \text{vol} = 0$). Whenever the probability density function ρ exists, we denote the expectation of φ as:

$$\langle \varphi, \rho \rangle := \int_{\mathcal{X}} \varphi(x) \rho(x) d \text{vol}^n$$

2 Fractal Dimension of Sets

Definition. The *diameter* of $A \subset \mathbb{R}^n$ is the maximum euclidean distance between any two points it contains.

$$\|A\| = \sup \{ \|x - y\|_2 : x, y \in A \}$$

Definition. The *s-dimensional Hausdorff measure* of $A \subset \mathbb{R}^n$ is defined as follows:

$$\mathcal{H}^s(A) = \lim_{\delta \rightarrow 0^+} \inf \left\{ \sum_{i=1}^{\infty} \|U_i\|^s : A \subseteq \bigcup_{i=1}^{\infty} U_i, 0 \leq \|U_i\| \leq \delta \right\}$$

Let's break this down. First, $\{U_i\}$ is what's called a δ -cover of A , meaning it is a set of sets whose union contains A (the first condition in the definition above) and whose diameter is at most δ (the second condition in the definition above). Then, the Hausdorff measure of A is the maximal sum of the s th powers of their diameters as the diameters go to zero. This definition is a strict generalization of volume, because one can show that the n -dimensional Hausdorff measure equals the volume measure up to a constant.

$$\mathcal{H}^n(A) = C_n^{-1} \text{vol}^n(A)$$

where C_n is the volume of an n -dimensional ball of diameter 1 (radius $1/2$).

$$C_n = \frac{\pi^{n/2}}{2^n \Gamma\left(\frac{n}{2} + 1\right)} = \begin{cases} \frac{\pi^{n/2}}{2^n \left(\frac{n}{2}\right)!}, & n \text{ even} \\ \frac{\pi^{(n-1)/2} \left(\frac{n-1}{2}\right)!}{n!}, & n \text{ odd} \end{cases}$$

When $m < n$, $\mathcal{H}^m$ gives the volume measure of lower dimensional subsets of $\mathbb{R}^n$, namely length ($m = 1$), area ($m = 2$), volume ($m = 3$) etc.

Definition. A consequence of the definition of the Hausdorff measure is that it must have a discontinuous jump at some critical value of s . To demonstrate this, consider the measure $\mathcal{H}_\delta^s$ before taking the $\delta \rightarrow 0^+$ limit. From our definition, we can derive the inequality $\mathcal{H}_\delta^t(A) \leq \delta^{t-s} \mathcal{H}_\delta^s(A)$. If there exists a value of s for which $\mathcal{H}^s(A) < \infty$ for the set A , then necessarily:

$$\mathcal{H}^t(A) = \lim_{\delta \rightarrow 0^+} \mathcal{H}_\delta^t \leq \lim_{\delta \rightarrow 0^+} \delta^{t-s} \mathcal{H}_\delta^s = 0, \quad \forall t > s$$

The consequence of this is there can be at most one value of s satisfying $0 < \mathcal{H}^s(A) < \infty$. We call this value the *Hausdorff dimension* of the set A :

$$\dim_H A = \inf \{s \geq 0 : \mathcal{H}^s(A) = 0\} = \sup \{s : \mathcal{H}^s(A) = \infty\}$$

Another way to put it is that:

$$\mathcal{H}^s(A) = \begin{cases} \infty, & 0 \leq s < \dim_H A \\ 0, & s > \dim_H A \end{cases}$$

With arbitrary (but not undefined) behavior at $s = \dim_H A$.

Definition. It should be apparent that the Hausdorff measure involves a very complex minimization over sets of subsets of $\mathbb{R}^n$. In particular, an optimal δ -cover $\{U_i\}$ generally involves sets at a wide range of diameters. However, if we think of δ as a kind of resolution, we typically just want to work with a global resolution variable. For that reason, we define *box-counting dimension* as follows:

$$\dim_B A = - \lim_{\delta \rightarrow 0^+} \frac{\log \mathcal{H}_\delta^0}{\log \delta} \quad (\text{when it exists})$$

While this is the most rigorous definition given how we have defined Hausdorff dimension thus far, the definition still suffers from the same problem I just described because even $\mathcal{H}_\delta^0$ (which assigns no penalty for covering set diameters below δ) is a minimum over sets of subsets with varying diameters. There is, however, a clever trick we can use to transform this definition into one with a global resolution. Consider the function $N_\delta(A)$ which counts the number of grid cells that $A \subset \mathbb{R}^n$ occupies (intersects) in a δ -resolution cubical grid over $\mathbb{R}^n$. Since the set of grid cells containing A is a cover of A , in particular a $(\delta\sqrt{n})$ -cover, the quantity $N_\delta(A)$ is just $\mathcal{H}_\delta^0$ before taking the infimum. So we have the following inequality:

$$N_\delta(A) = \sum_{i=1}^{\infty} \|U_i^{\delta\text{-grid}} \cap A\|^0 \geq \inf \left\{ \sum_{i=1}^{\infty} \|U_i\|^0 : \{U_i\} \text{ is a } (\delta\sqrt{n})\text{-cover} \right\} = \mathcal{H}_{\delta\sqrt{n}}^0(A)$$

Similarly, we can derive an upper bound for $N_\delta(A)$ in terms of $\mathcal{H}_\delta^0$. If we take the minimal δ -cover $\{U_i\}^*$ which defines $\mathcal{H}_\delta^0$, then count the δ -grid cell which it must intersect plus the neighboring $3^n - 1$ cells which it may intersect, this is evidently going to overcount the number of individual cells intersecting A . A suitable upper bound is therefore:

$$N_\delta(A) \leq 3^n \mathcal{H}_\delta^0(A)$$

Now we have two bounds, and when we take logarithms and evaluate the $\delta \rightarrow 0^+$ limit, the two bounds become identical, sandwiching the middle quantity:

$$\begin{aligned} \mathcal{H}_{\delta\sqrt{n}}^0(A) &\leq N_\delta(A) \leq 3^n \mathcal{H}_\delta^0(A) \\ \frac{\log \mathcal{H}_{\delta\sqrt{n}}^0(A)}{-\log \delta} &\leq \frac{\log N_\delta(A)}{-\log \delta} \leq \frac{\log(3^n \mathcal{H}_\delta^0(A))}{-\log \delta} \\ \lim_{\delta \rightarrow 0^+} \frac{\log \mathcal{H}_{\delta\sqrt{n}}^0(A)}{-\log \delta} &\leq \lim_{\delta \rightarrow 0^+} \frac{\log N_\delta(A)}{-\log \delta} \leq \lim_{\delta \rightarrow 0^+} \frac{\log(3^n \mathcal{H}_\delta^0(A))}{-\log \delta} \\ \lim_{\delta\sqrt{n} \rightarrow 0^+} \frac{\log \mathcal{H}_{\delta\sqrt{n}}^0(A)}{-\log(\delta\sqrt{n}) + \log \sqrt{n}} &\leq \lim_{\delta \rightarrow 0^+} \frac{\log N_\delta(A)}{-\log \delta} \leq \lim_{\delta \rightarrow 0^+} \frac{\log \mathcal{H}_\delta^0(A) + n \log 3}{-\log \delta} \\ \lim_{\delta \rightarrow 0^+} \frac{\log \mathcal{H}_\delta^0(A)}{-\log \delta} &\leq \lim_{\delta \rightarrow 0^+} \frac{\log N_\delta(A)}{-\log \delta} \leq \lim_{\delta \rightarrow 0^+} \frac{\log \mathcal{H}_\delta^0(A)}{-\log \delta} \end{aligned}$$

Therefore,

$$\dim_B A = - \lim_{\delta \rightarrow 0^+} \frac{\log \mathcal{H}_\delta^0(A)}{\log \delta} = - \lim_{\delta \rightarrow 0^+} \frac{\log N_\delta(A)}{\log \delta}$$

and finally, we see exactly why this is called the box-counting dimension. With some admittedly technically-involved manipulations, we have shown that box-counting dimension as defined via the 0-dimensional Hausdorff measure is exactly the same as the asymptotic log-log ratio of the number of grid cells need to cover a set at a given resolution to the resolution itself. Instead of an intractable minimization problem required to compute dimension,

we have reduced it to a simple counting problem on a discretized grid. That's why this is the definition used to estimate the fractal dimension of coastlines, for example. The major drawback is that this is of course no longer the same quantity as the Hausdorff dimension. In fact, we cannot even say anything about how much it deviates from the Hausdorff dimension.

In practice, choosing the allowable set of δ -covers is equivalent to choosing a compression scheme, and the Hausdorff dimension allows compression schemes that are so complex so as to be meaningless. For example, box-counting dimension can answer the question “*How many bits do I need to encode a fractal set at a given resolution?*”, but it's not so clear what kind of question the Hausdorff dimension answers. Theoretically, the only universal relation we can do is the following inequality, which is intuitively a statement about the fact that box-counting is not necessarily the most efficient way to cover a set.

$$\dim_H A \leq \dim_B A$$

Box-counting dimension also has a convergence problem. The limit $\delta \rightarrow 0^+$ may only exist as an upper/lower bound, so the box-counting dimension as a single scalar does not necessarily exist. For that reason, you may see the following definitions for *upper* and *lower box-counting dimension*:

$$\begin{aligned}\overline{\dim}_B A &= -\limsup_{\delta \rightarrow 0^+} \frac{\log N_\delta(A)}{\log \delta} \\ \underline{\dim}_B A &= -\liminf_{\delta \rightarrow 0^+} \frac{\log N_\delta(A)}{\log \delta}\end{aligned}$$

One final note on box-counting dimension is that there are many variations on the definition of N_δ which all give the same box-counting dimension. To list a few examples, we could instead count the smallest number of arbitrarily oriented cubes of side length δ needed to cover A , or the smallest number of balls of radius δ needed to cover A , or the even the largest number of disjoint balls of radius δ whose centers are in A . The overarching theme is that when grouping elements of A by some small-resolution clustering scheme, the number of clusters required tends to grow with the power law:

$$N_\delta(A) \sim C \delta^{-s}$$

Next, we'll look at one final equivalent definition of box-counting dimension which will...

Definition. The δ -neighbourhood of a set $A \subset \mathbb{R}^n$ is the set of points within δ distance from any point in A :

$$A_\delta = \{x \in \mathbb{R}^n : \|x - y\|_2 \leq \delta, y \in A\}$$

By a similar kind of argument to the one we used to show the equivalence of box-counting dimension under different counting schemes, we can show that the box-counting dimension is also equivalent to the following limit:

$$\dim_B A = n - \lim_{\delta \rightarrow 0^+} \frac{\log \text{vol}^n(A_\delta)}{\log \delta}$$

3 Fractal Dimension of Measures

Definition. The *Rényi dimension* is defined as follows:

$$\mathcal{D}_q[\mu] = \lim_{\delta \rightarrow 0} \frac{1}{q-1} \frac{\log \sum_i \mu(U_i^\delta)^q}{\log \delta}$$

where $\{U_i^\delta\}$ is a δ -resolution grid over $\mathbb{R}^n$. This is considered a strict generalization of the box-counting dimension because it is exactly equal at $q = 0$:

$$\mathcal{D}_0[\mu] = \lim_{\delta \rightarrow 0} \frac{\log \sum_i \mu(U_i^\delta)^0}{-\log \delta} = \lim_{\delta \rightarrow 0} \frac{\log N_\delta(\text{supp } \mu)}{-\log \delta} = \dim_B \text{supp } \mu$$

For $q = 2$, the Rényi dimension is equal to what's called the correlation dimension, which is the dimension estimate obtained by computing the average distance of pairs of nearby points. One might expect $\mathcal{D}_1$ to not exist because it results in a denominator of zero, but taking the limit as $q \rightarrow 1$ gives a special convergent expression:

$$\mathcal{D}_1[\mu] = \lim_{\delta \rightarrow 0} \lim_{q \rightarrow 1} \frac{1}{q-1} \frac{\log \sum_i \mu(U_i^\delta)^q}{\log \delta} = \lim_{\delta \rightarrow 0} \frac{\sum_i \mu(U_i^\delta) \log \mu(U_i^\delta)}{\log \delta} = \lim_{\delta \rightarrow 0} \frac{H_\delta[\mu]}{\log \delta}$$

where $H_\delta[\mu]$ is the so-called Shannon block entropy of the distribution μ at resolution δ . This is called the *information dimension*. Just like the box-counting dimension, there are many equivalent definitions of the information dimension which define finite-resolution entropy in different ways. In their 2007 paper titled “On Classical Analogues of Free Entropy Dimension”, Giuonnet and Shlyakhtenko demonstrated that the following is true for any probability measures μ, ν with $H[\nu] < \infty$:

$$\mathcal{D}_1[\mu] = n - \lim_{\sigma \rightarrow 0} \frac{H[\mu * \nu_\sigma]}{\log \sigma} = n - \lim_{\sigma \rightarrow 0} \frac{H(X + \sigma Y)}{\log \sigma}, \quad X \sim \mu, Y \sim \nu$$

Here, ν_σ is the pushforward of the measure ν by the dilation operator $\Phi_\sigma : x \mapsto \sigma x$ and $H[\nu] = -\int_{\mathbb{R}^n} \rho(x) \log \rho(x) \text{vol}^n$ is the differential entropy of the probability density function. This is a very powerful statement because the only restriction on ν is that it has finite entropy and that both measures have finite log-moment $\int \log(1 + |x|) d\mu < \infty$. This invariance under choices of ν means that information dimension tells us something about the fundamental scaling properties of the measure—very interesting! Next we will explore the most useful choice of ν_σ , which is the Gaussian distribution.

4 Gaussian Scale

Definition. The *Gaussian measure* ν_σ (often denoted $\mathcal{N}(0, \sigma^2 I)$) is the probability measure satisfying:

$$\nu_\sigma(A) = \int_A g_\sigma(x) \text{vol}_x^n$$

where $g_\sigma(x)$ is the Gaussian probability density function,

$$g_\sigma(x) = \frac{1}{(\sqrt{2\pi}\sigma)^n} \exp\left(-\frac{\|x\|_2^2}{2\sigma^2}\right)$$

which is, importantly, a dilation of the standard normal distribution $\mathcal{N}(0, I)$:

$$g_\sigma(x) = \frac{1}{\sigma^n} g_1\left(\frac{x}{\sigma}\right)$$

The σ^{-n} factor comes from the pushforward of the volume measure since all n dimensions are being dilated by σ .

Definition. A *Gaussian-smoothed measure* μ_σ is the convolution of a probability measure μ with the Gaussian measure ν_σ :

$$\begin{aligned} \mu_\sigma(A) &= (\mu * \nu_\sigma)(A) = \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \mathbf{1}_A(x+y) g_\sigma(y) \text{vol}_y^n d\mu_x \\ &= \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \mathbf{1}_A(z) g_\sigma(z-x) \text{vol}_z^n d\mu_x \\ &= \int_{\mathbb{R}^n} \mathbf{1}_A(z) \left(\int_{\mathbb{R}^n} g_\sigma(z-x) d\mu_x \right) \text{vol}_z^n \\ &= \int_A \rho_\sigma \text{vol}^n \end{aligned}$$

From the definition, we see that a gaussian smoothed measure always has a probability density function,

$$\rho_\sigma(x) = \int_{\mathbb{R}^n} g_\sigma(x-y) d\mu_y$$

which we can now use to compute the information dimension of μ :

$$\mathcal{D}_1[\mu] = n + \lim_{\sigma \rightarrow 0} \frac{H[\mu_\sigma]}{\log \sigma} = n - \lim_{\sigma \rightarrow 0} \frac{\int_{\mathbb{R}^n} \rho_\sigma(x) \log \rho_\sigma(x) \text{vol}_x^n}{\log \sigma}$$

An important property of ρ_σ is that it satisfies a heat equation in the variance $\sigma^2/2$:

$$\begin{aligned} \partial_\sigma \rho_\sigma(x) &= \frac{\partial}{\partial \sigma} \int_{\mathbb{R}^n} g_\sigma(x-y) d\mu_y \\ &= \int_{\mathbb{R}^n} \frac{\partial}{\partial \sigma} \left(\frac{1}{\sigma^n} g_1\left(\frac{x-y}{\sigma}\right) \right) d\mu_y \\ &= - \int_{\mathbb{R}^n} \frac{n}{\sigma^{n+1}} g_1\left(\frac{x-y}{\sigma}\right) + \frac{x-y}{\sigma^{n+3}} \cdot \nabla g_1\left(\frac{x-y}{\sigma}\right) d\mu_y \\ &= - \int_{\mathbb{R}^n} \frac{n}{\sigma} g_\sigma(x-y) + \frac{x-y}{\sigma^3} \cdot \nabla g_\sigma(x-y) d\mu_y \\ &= \int_{\mathbb{R}^n} \sigma (\nabla \cdot \nabla \log g_\sigma(x-y)) g_\sigma(x-y) + \sigma \nabla \log g_\sigma(x-y) \cdot \nabla g_\sigma(x-y) d\mu_y \\ &= \int_{\mathbb{R}^n} \sigma \nabla \cdot (g_\sigma(x-y) \nabla \log g_\sigma(x-y)) d\mu_y \\ &= \int_{\mathbb{R}^n} \sigma \Delta g_\sigma(x-y) d\mu_y \\ &= \sigma \Delta \rho_\sigma(x) \end{aligned}$$

Using $\frac{1}{\sigma}\partial_\sigma = \partial_{\sigma^2/2}$, we recover a heat equation in the variance. Now we can derive the *deBruijn identity* which relates the derivative of entropy with respect to scale to a quantity called the *Fisher information* $\mathcal{J}$ as follows:

$$\begin{aligned}
\partial_\sigma H[\mu_\sigma] &= -\partial_\sigma \int_{\mathbb{R}^n} \rho_\sigma(x) \log \rho_\sigma(x) \text{vol}_x^n \\
&= -\int_{\mathbb{R}^n} \partial_\sigma \rho_\sigma(x) \log \rho_\sigma(x) + \partial_\sigma \rho_\sigma(x) \text{vol}_x^n \\
&= -\sigma \int_{\mathbb{R}^n} \Delta \rho_\sigma(x) (\log \rho_\sigma(x) + 1) \text{vol}_x^n \\
&= -\sigma \int_{\mathbb{R}^n} \nabla \cdot ((\log \rho_\sigma(x) + 1) \nabla \rho_\sigma(x)) - \nabla \log \rho_\sigma(x) \cdot \nabla \rho_\sigma(x) \text{vol}_x^n \\
&= -\sigma \underbrace{\int_{\partial \mathbb{R}^n} (\log \rho_\sigma(x) + 1) \nabla \rho_\sigma(x) \cdot \hat{n} \text{vol}_x^{n-1}}_{\text{boundary flux}} + \sigma \underbrace{\int_{\mathbb{R}^n} \|\nabla \log \rho_\sigma(x)\|_2^2 \text{d}\mu_\sigma}_{\text{Fisher information } \mathcal{J}[\mu_\sigma]}
\end{aligned}$$

Typically, we assume the flux at the boundary of $\mathbb{R}^n$ is zero, so the deBruijn identity is any of the following:

$$\begin{aligned}
\partial_\sigma H[\mu_\sigma] &= \sigma \mathcal{J}[\mu_\sigma] && \text{(standard deviation)} \\
\partial_\tau H[\mu_\tau] &= \tfrac{1}{2} \mathcal{J}[\mu_\tau] && \text{(variance } \tau = \sigma^2) \\
\partial_s H[\mu_s] &= \mathcal{J}[\mu_s] && \text{(heat time } s = \sigma^2/2) \\
\partial_\lambda H[\mu_\lambda] &= e^{2\lambda} \mathcal{J}[\mu_\lambda] && \text{(log scale } \tau = \log \sigma)
\end{aligned}$$

We can now use the fundamental theorem of calculus to write entropy as an integral:

$$H[\mu_\sigma] = H[\mu_1] + \int_1^\sigma \mathcal{J}[\mu_s] \text{d}s$$

giving a new expression for the information dimension which averages the Fisher information over all $s \leq 0$:

$$\mathcal{D}_1[\mu] = n - \lim_{\sigma \rightarrow 0} \frac{1}{\log \sigma} \int_\sigma^1 \mathcal{J}[\mu_s] \text{d}s$$

5 Dynamical Systems

Definition. A *finite dimensional dynamical system* consists of a state space $\mathcal{X} \subseteq \mathbb{R}^n$ and a velocity field $v : [0, \infty) \times \mathcal{X} \rightarrow \mathbb{R}^n$ which defines a system of n ordinary differential equations:

$$\begin{aligned}
\partial_t x^i(t) &= v^i(x, t), \quad i = 1, \dots, n \\
x^i(0) &= x_0^i
\end{aligned}$$

If the velocity field is independent of time, we call it an *autonomous* dynamical system. We will only consider autonomous dynamical systems from now on.

Definition. The *flow* generated by a dynamical system is 1-dimensional family of diffeomorphisms $\{\Phi_t\} : \mathcal{X} \rightarrow \mathcal{X}$ satisfying the following properties:

1. $\Phi_0 = \text{id}$ (identity element)
2. $\Phi_{t+s} = \Phi_t \circ \Phi_s$ for all $t, s \in \mathbb{R}^+$ (group property)
3. $\partial_t \Phi_t(x)|_{t=0} = v(x)$ for all $x \in \mathcal{X}$ (initial condition)

From these properties, one can show that $\partial_t \Phi_t(x) = v(\Phi_t(x))$, $\forall t \geq 0, x \in \mathcal{X}$,

$$\begin{aligned} v(\Phi_t(x)) &= \partial_s \Phi_s(\Phi_t(x))|_{s=0} \\ &= \partial_s \Phi_{s+t}(x)|_{s=0} \\ &= \partial_t \Phi_{s+t}(x)|_{s=0} \\ &= \partial_t \Phi_t(x) \end{aligned}$$

and therefore Φ_t is the unique solution to the system of ODEs defined by v :

$$\begin{aligned} \partial_t x(t) &= v(x(t)) \\ x(0) &= x_0 \\ \implies x(t) &= \Phi_t(x_0) \end{aligned}$$

Definition. The *pushforward measure* μ_t of a dynamical system is the probability measure which satisfies the following property for all smooth functions $\varphi : \mathcal{X} \rightarrow \mathbb{R}$:

$$\int_{\mathcal{X}} \varphi d\mu_t = \int_{\Phi_t(\mathcal{X})} \varphi d\mu, \quad \forall t \geq 0$$

Equivalently, it is the measure which satisfies the following property for all $A \subseteq \mathcal{X}$:

$$\mu_t(\{\Phi_t(x) : x \in A\}) = \mu_t(\Phi_t(A)) = \mu(A)$$

Upon applying the group property of the flow, we get the more general conditions:

$$\begin{aligned} \int_{\mathcal{X}} \varphi(\Phi_{t+\tau}(x)) d\mu &= \int_{\mathcal{X}} \varphi(\Phi_t(\Phi_\tau(x))) d\mu = \int_{\mathcal{X}} \varphi(\Phi_\tau(x)) d\mu_t \\ \mu_{t+\tau}(\Phi_{t+\tau}(A)) &= \mu_{t+\tau}(\Phi_t(\Phi_\tau(A))) = \mu_\tau(\Phi_\tau(A)) = \mu(A) \end{aligned}$$

Intuitively, the pushforward measure deterministically evolves uncertainty in the initial state while ensuring conservation of probability. Let's now assume that the pushforward measure has a density $\rho(x, t)$, with the initial condition $\rho(x, 0) = \rho_0(x)$ where $\rho_0(x)$ is the density of the initial measure μ . The identity we require is:

$$\int_{\mathcal{X}} \varphi(x) \rho(x, t + \tau) d \text{vol}^n = \int_{\mathcal{X}} \varphi(\Phi_\tau(x)) \rho(x, t) d \text{vol}^n$$

Differentiating the identity above about $\tau = 0$, we get:

$$\begin{aligned} \left. \partial_\tau \int_{\mathcal{X}} \varphi(x) \rho(x, t + \tau) \, d \text{vol}^n \right|_{\tau=0} &= \left. \partial_\tau \int_{\mathcal{X}} \varphi(\Phi_\tau(x)) \rho(x, t) \, d \text{vol}^n \right|_{\tau=0} \\ \left. \partial_t \int_{\mathcal{X}} \varphi(x) \rho(x, \tau + t) \, d \text{vol}^n \right|_{\tau=0} &= \int_{\mathcal{X}} v(\Phi_\tau(x)) \cdot \nabla \varphi(\Phi_\tau(x)) \rho(x, t) \, d \text{vol}^n \Big|_{\tau=0} \\ \partial_t \int_{\mathcal{X}} \varphi(x) \rho(x, t) \, d \text{vol}^n &= \int_{\mathcal{X}} v(x) \cdot \nabla \varphi(x) \rho(x, t) \, d \text{vol}^n \end{aligned}$$

Assume that $(\varphi \rho v)$ vanishes at the boundary. Then, by the divergence theorem, we have:

$$\begin{aligned} \int_{\mathcal{X}} \nabla \cdot (\varphi(x) \rho(x, t) v(x)) \, d \text{vol}^n &= \int_{\partial \mathcal{X}} \varphi(x) \rho(x, t) v(x) \cdot \hat{n} \, d \text{vol}^{n-1} = 0 \\ \implies \int_{\mathcal{X}} v(x) \cdot \nabla \varphi(x) \rho(x, t) \, d \text{vol}^n &= - \int_{\mathcal{X}} \varphi(x) \nabla \cdot (\rho(x, t) v(x)) \, d \text{vol}^n \end{aligned}$$

The only way to enforce equality for all φ is to impose the following PDE on ρ :

$$\begin{aligned} \partial_t \rho(x, t) &= -\nabla \cdot (\rho(x, t) v(x)) \\ \rho(x, 0) &= \rho_0(x) \end{aligned}$$

This is called the *continuity equation* or *Liouville equation*. Using the continuity equation, we can derive a general formula for expectations in terms of the coordinates $x = \Phi_t(x_0)$ at any future time t . A somewhat tedious calculation making use of Jacobi's formula for the derivative of the determinant of a matrix gives us a closed form solution to the evolution of the density along trajectories:

$$\begin{aligned} \partial_t \rho(\Phi_t(x_0), t) &= \left[v(x) \cdot \nabla \rho(x, t) + \partial_t \rho(x, t) \right]_{x=\Phi_t(x_0)} \\ \partial_t \rho(\Phi_t(x_0), t) &= \left[v(x) \cdot \nabla \rho(x, t) - \nabla \cdot (\rho(x, t) v(x)) \right]_{x=\Phi_t(x_0)} \\ &= \left[v(x) \cdot \nabla \rho(x, t) - v(x) \cdot \nabla \rho(x, t) - \rho(x, t) \nabla \cdot v(x) \right]_{x=\Phi_t(x_0)} \\ \partial_t \rho(\Phi_t(x_0), t) &= \left[-\rho(x, t) \nabla \cdot v(x) \right]_{x=\Phi_t(x_0)} \\ \partial_t \log \rho(\Phi_t(x_0), t) &= - \left[\text{Tr}(\nabla v(x)) \right]_{x=\Phi_t(x_0)} \\ \partial_t \log \rho(\Phi_t(x_0), t) &= - \text{Tr}((\nabla \Phi_t(x_0))^{-1} \partial_t (\nabla \Phi_t(x_0))) \\ &= -\partial_t \log \det \nabla \Phi_t(x_0) \quad \text{(by Jacobi's formula)} \\ \implies \rho(\Phi_t(x_0), t) &= \frac{\rho(x_0, 0)}{\det \nabla \Phi_t(x_0)} \end{aligned}$$

The determinant of the flow jacobian can therefore be interpreted as a corrective factor for the volume measure upon mapping points along the flow, so we can write,

$$d \text{vol}_{\Phi_t(x)}^n = \det \nabla \Phi_t(x_0) \, d \text{vol}_{x_0}^n$$

as a notational trick. This is exactly the change of variables formula for integrals! Except now all the measure theoretic inner workings are laid out before you and you can finally grasp the true nature of integration.

$$\langle \varphi, \rho \rangle(t) = \int_{\mathcal{X}} \varphi(\Phi_t(x_0)) \rho(x_0, 0) \, d \operatorname{vol}_{x_0}^n = \int_{\mathcal{X}} \varphi(\Phi_t(x_0)) \rho(\Phi_t(x_0), t) \, d \operatorname{vol}_{x(t)}^n = \int_{\mathcal{X}} \varphi(x) \rho(x, t) \, d \operatorname{vol}_x^n$$

6 Fractal Attractors

One immediate consequence of the continuity equation is that the density grows exponentially with the divergence of the velocity field along trajectories:

$$\begin{aligned} \partial_t \rho(x, t) &= -\nabla \cdot (\rho(x, t) v(x)) = -v(x) \cdot \nabla \rho(x, t) - \rho(x, t) \nabla \cdot v(x) \\ \implies \frac{d}{dt} \rho(x(t), t) &= -(\nabla \cdot v(x(t))) \rho(x(t), t) \end{aligned}$$

Of course if the divergence of the velocity field is zero, the density is constant along trajectories, but if the divergence takes a nonzero average value Λ then the density grows like $\rho \sim e^{-\Lambda t}$. Suppose $\Lambda < 0$. This means the density has a singularity at $t = \infty$, which should feel very wrong, because how can probability be conserved if it's growing exponentially everywhere? The answer is that the probability mass is concentrating on smaller and smaller sets culminating in a reduction of dimension at the singularity.

Consider the simplest possible example: $v(x) = -x$. The divergence is just $-n$, so we know that the pushforward measure's probability density must grow pointwise like $\rho = Ce^t$. It's also easy to determine that the flow is simply $\Phi_t(x) = e^{-t}x$, therefore $\lim_{t \rightarrow \infty} \Phi_t(x) = \vec{0}$. The only probability measure whose support is the origin is the Dirac delta measure δ_0 which has the property that $\int \varphi(x) d\delta_0 = \varphi(0)$. Therefore, the pushforward measure at $t = \infty$ is δ_0 , which has a dimension of zero regardless of how you define dimension. You could calculate the Hausdorff dimension of the support, or the box counting dimension of the support, or even evaluate the limit $\mathcal{D}_I[\delta_0] = n + \lim_{\sigma \rightarrow 0} H[\sigma Y] / \log \sigma$ because the entropy of the gaussian distribution scales like $H[\sigma Y] \sim -n \log \sigma$. All of these would yield a dimension of zero. That's because the Dirac delta measure is really just the zero dimensional volume measure vol^0 , but it's almost always presented as a distribution over $\mathbb{R}^n$.

Now it should be somewhat less surprising to learn that certain kinds of dynamical systems can result in fractal-dimensional pushforward measures. We'll need to find a set of conditions more precise than $\Lambda < 0$ to formally define what it means for a dynamical system to appear fractal-dimensional.

Definition. The *limit measure* μ_∞ of a dynamical system is simply the pushforward measure at $t = \infty$:

$$\mu_\infty = \lim_{t \rightarrow \infty} \mu_t$$

If this limit exists, like it did in our simple example, then the analysis of long term dynamical behavior is simplified a great deal. Unfortunately, there are many reasons why this limit may not exist, and it is generally difficult to characterize the conditions under which it does. For

that reason, we will have to loosen the limit measure notion to an attractor set and impose some conditions on the flow to ensure that the pushforward measure concentrates on this set.

Definition. A dynamical system has a *global attractor* A^* if there exists a bounded set $B \subset \mathbb{R}^n$ such that every point in $\mathbb{R}^n$ enters B in finite time. In that case, the global attractor is defined as the minimal set containing all points reached and approached by trajectories starting in B . This is defined using something called an ω -limit set,

$$A^* = \omega(B) := \bigcap_{t \geq 0} \overline{\bigcup_{s \geq t} \Phi_s(B)}$$

which is an esoteric way of formalizing what I said in words. Importantly, the over-bar on the union is called a closure in topology and it is what lets us include the limiting points as well as the points reached by trajectories.

The existence of a global attractor is a very powerful statement with a relatively simple existence condition. It tells us that regardless of whether the limit measure exists, the support of the pushforward measure does converge to exactly A^* regardless of the initial measure. In other words, the distribution of probability mass on the attractor may evolve forever, but the set of states over which it is distributed converges to a fixed set. Furthermore, it guarantees at least one invariant measure.

7 Making Fractal Dimension Dynamical

For a dynamical system with initially Lebesgue-continuous measure, we have $\mathcal{D}_1[\mu_0] = n$. Now suppose $0 < \mathcal{D}_1[\mu_\infty] < n$. The question is: what happens during the transient? Given our existing definitions, imposing a continuity equation on the measure ensures that it is smooth for all finite times, and any smooth measure must have a full dimension of n . This is very problematic because that means information dimension is not a function of time over the real numbers. In particular, if fractal dimension is a function of $\mathcal{D}_1(t)$, it would have the properties,

$$\begin{aligned} \mathcal{D}_1(t) &= n, \quad \forall t \in [0, \infty) \\ \lim_{t \rightarrow \infty} \mathcal{D}_1(t) &= \mathcal{D}_1[\mu_\infty] < n \end{aligned}$$

which are contradictory. The problem is that for finite time, the limits are interchangeable, but at infinite time, the limits are not interchangeable. Namely,

$$\mathcal{D}_I[\mu_\infty] = \mathcal{D}_1 \left[\lim_{t \rightarrow \infty} \mu_t \right] \neq \lim_{t \rightarrow \infty} \mathcal{D}_1[\mu_t]$$